

Embeddings and Semantic Similarity

An embedding represents a word, sentence, or longer passage as a vector of numbers. The individual numbers are not intended to be interpreted directly by a human. What matters is that texts with similar meanings are usually placed near each other in the embedding space.

For example, the questions "Where is the database stored?" and "Which file contains the data?" may use different words but still express similar intent. A useful embedding model can map these sentences to nearby regions of the vector space.

In a RAG application, document chunk embeddings are normally created once when a document is added to the knowledge base. If the document has not changed, its embeddings do not need to be recomputed for every question. At query time, only the user question needs a new embedding.

Cosine similarity measures how similar the directions of two vectors are. A higher score usually indicates stronger semantic similarity between two texts. However, the score is not a correctness percentage. For example, a cosine similarity score of 0.72 does not mean that an answer is 72 percent correct.

During retrieval, the system can rank all stored chunks by similarity score and select the highest-ranked results. The number of selected chunks is commonly called top-K. If K is too small, useful context may be missed. If K is too large, irrelevant context may be sent to the language model.